

GREEN CLOUD FRAMEWORK FOR AI-AWARE VIRTUAL MACHINE PLACEMENT AND ENERGY OPTIMIZATION

A CAPSTONE PROJECT REPORT

*A Capstone Project Report submitted in partial fulfilment of the requirements
for the Course of*

**Course Code & Course Name: CSA1501-CLOUD COMPUTING AND BIG
DATA ANALYTICS**

to the award of the degree of

BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE AND ENGINEERING

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CERTIFICATE FROM THE SUPERVISOR

This is to certify that the Capstone Project entitled “Green Cloud Framework for AI-Aware Virtual Machine Placement and Energy Optimization” has been carried out by Deepak R (192511008) and Bharath S (192521034) under the academic guidance of the course faculty and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech Computer Science and Engineering programme at Saveetha Institute of Medical and Technical Sciences, Chennai.

The work presented in this report describes a CloudSim Plus based simulation framework for evaluating virtualized cloud infrastructure, monitoring resource utilization, estimating energy consumption, observing VM migration behaviour, and presenting simulation results through a live dashboard and exported reports.

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DECLARATION BY THE CANDIDATE

We, Deepak R (192511008) and Bharath S (192521034), students of the Computer Science and Engineering Department, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “Green Cloud Framework for AI-Aware Virtual Machine Placement and Energy Optimization” is the result of our academic project work.

To the best of our knowledge, the work presented in this report is original and has been carried out in accordance with engineering ethics, academic integrity, and the requirements of the capstone project. All external literature, software frameworks, libraries, and technical resources used in the development have been appropriately acknowledged.

The project uses CloudSim Plus as a simulation environment rather than a physical cloud data centre. The numerical outputs reported in this document therefore represent simulated infrastructure behaviour under the configurations specified in the test cases. The project does not claim that the measured energy values are direct measurements from physical servers.

The dashboard, monitoring components, CSV exports, test configurations, graphs, and analysis presented in this report correspond to the implementation developed for this project. Where external research is discussed, it is used to establish the engineering context and identify the gap addressed by the proposed simulation and monitoring framework.

We further declare that the results, observations, and conclusions reported here have not been intentionally fabricated, falsified, or manipulated. Any AI-assisted development or coding support used during implementation was treated as a development aid, while the final integration, testing, execution, and interpretation were performed for this project.

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Date: September 2026

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INDIVIDUAL CONTRIBUTION STATEMENT

(Mandatory for all team projects)

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
Deepak R 192511008	Simulation architecture, CloudSim setup, Java/Maven integration, monitoring dashboard, CSV reporting, system integration	CloudSim setup, managers, monitoring and dashboard implementation	Test execution, debugging, output validation and graph analysis	Architecture, implementation, testing, results and documentation	50%
Bharath S 192521034	Project analysis, experiment configuration, validation support and documentation	Configuration design, experiment preparation and supporting implementation	Test-case comparison, result checking and interpretation	Literature review, analysis, conclusion and supporting sections	50%
Total					100%

The contribution split is intended as an academic reporting representation of the overall team effort. Individual tasks were not isolated completely; both members participated in discussions, integration decisions, verification, and preparation of the final demonstration.

The implementation followed a modular approach. Core simulation functionality was kept separate from monitoring and visualization so that changes in presentation did not alter the underlying CloudSim simulation behaviour. This separation also helped during debugging of dashboard layout, utilization snapshots, and report refresh behaviour.

ABSTRACT

Cloud data centres provide scalable computing resources through virtualization, but inefficient resource utilization can increase energy consumption and reduce infrastructure efficiency. Evaluating infrastructure decisions directly on a physical cloud environment is costly and difficult during early design stages. This project therefore develops a simulation-based Green Cloud Framework that allows cloud infrastructure configurations to be studied before physical deployment.

The proposed system is implemented in Java using Maven and CloudSim Plus. A configurable simulation launcher accepts the number of hosts, number of virtual machines, cloudlet workload duration, host CPU capacity, and VM CPU requirement. The simulation creates the virtual data-centre environment and executes workloads while a monitoring subsystem records host, VM, cloudlet, energy, migration, and SLA-related metrics. A live dashboard presents key performance indicators, time-series graphs, host summaries, and migration information. CSV reports are also generated for later analysis.

Three representative test configurations were evaluated. Case 1 used 10 hosts and 20 VMs and produced 3.5933 Wh of simulated energy, 11.25% average host CPU utilization, 45.00% average VM CPU utilization, 20 migrations, and zero SLA violations. Case 2 used 50 hosts and 150 VMs and produced 45.9306 Wh, 4.08% host CPU utilization, 16.33% VM CPU utilization, and 150 migrations with zero SLA violations. Case 3 used 50 hosts and 200 VMs and produced a dashboard snapshot of 43.4115 Wh, 3.06% host CPU utilization, 12.25% VM CPU utilization, 200 migrations, and 3450 reported SLA violations during the captured running state.

The project demonstrates that CloudSim Plus can be used as a safe experimental environment for studying infrastructure behaviour, monitoring resource utilization, and comparing workload configurations without requiring a physical cloud deployment. The framework also provides a foundation for future AI-aware placement and energy-optimization algorithms.

Keywords: Green Cloud, CloudSim Plus, Virtual Machine Placement, Energy Monitoring, VM Migration, Cloud Simulation

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The architecture and workflow diagrams were prepared specifically for this report to communicate the relationship between the simulation layer, monitoring subsystem, energy model, dashboard, and exported analytics. The result graphs summarize the experimental outputs recorded from the project test cases.

The live dashboard is designed around KPI cards, time-series views, migration information, and host-level summaries. The original report describes the dashboard structure but does not contain a separate dashboard screenshot; an editable placeholder is therefore retained in Chapter 4 for insertion if required.

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol / Term	Description	Unit / Context
AI	Artificial Intelligence	Computing method
CPU	Central Processing Unit	MIPS / %
CSV	Comma-Separated Values	Report format
DC	Data Centre / Datacenter	Infrastructure
GUI	Graphical User Interface	Application UI
MIPS	Million Instructions Per Second	CPU capacity
RAM	Random Access Memory	% / GB
SLA	Service Level Agreement	Constraint
VM	Virtual Machine	Virtualized resource
Wh	Watt-hour	Energy
W	Watt	Power
VMP	Virtual Machine Placement	Resource management
QoS	Quality of Service	Performance constraint
Maven	Java build and dependency management tool	Build system
CloudSim Plus	Cloud computing simulation framework	Simulation
SLF4J	Simple Logging Facade for Java	Logging API
Logback	Logging implementation used with SLF4J	Logging

In this project, CPU utilization is expressed as a percentage of available capacity, while MIPS values describe simulated processing capacity. Energy is accumulated in watt-hours based on the selected simulation energy model.

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Cloud computing relies heavily on virtualization to share physical infrastructure among multiple workloads. A physical host can execute several virtual machines, while each VM can process one or more cloudlets representing application workloads. This architecture provides flexibility and scalability, but it also creates a resource-management problem: the placement of VMs across hosts influences CPU utilization, memory utilization, power consumption, migration activity, and service quality.

A conventional infrastructure design process may require significant investment before the behaviour of a chosen configuration can be understood. For an academic or engineering team, experimenting directly with a production cloud is impractical because it introduces cost, operational risk, and limited control over repeatability. Simulation provides a controlled alternative in which the number of hosts, VMs, workload length, CPU capacities, and other conditions can be changed systematically.

The Green Cloud Framework developed in this project treats the simulated data centre as an experimental laboratory. Instead of simply running a workload and observing whether it completes, the system records infrastructure-level information and presents it through a monitoring dashboard. This makes it possible to study how workload scale changes utilization and energy behaviour.

The framework is implemented using Java 17, Maven, and CloudSim Plus. The project is organized into simulation, monitoring, energy, and user-interface responsibilities. The monitoring layer collects snapshots during execution, while the dashboard converts these snapshots into human-readable KPIs, time-series plots, migration information, and host-level summaries.

The engineering significance is therefore not limited to the graphical interface. The core value is the ability to create controlled cloud scenarios, execute them repeatedly, collect comparable metrics, and use those results to support infrastructure decisions before physical deployment.

1.1.1 Application Context

The intended application context includes academic cloud research, infrastructure prototyping, workload planning, energy-efficiency studies, and demonstrations of VM management concepts. The framework can be used to answer questions such as how many hosts are required for a workload, how VM density changes utilization, how much simulated energy is accumulated, and how migration behaviour changes as the system scales.

1.2 Need for the Project

Energy efficiency is an important engineering consideration in cloud data centres because physical infrastructure consumes power even when resource utilization is low. Virtualization improves sharing of resources, but poor placement can leave hosts underutilized or create overloaded conditions.

Engineers therefore need a way to study resource-management strategies before committing to physical infrastructure.

The project is particularly useful where multiple interacting parameters must be explored. Host count, VM count, CPU capacity, workload duration, migrations, utilization and SLA behaviour are not independent variables. Changing one configuration can affect several metrics simultaneously. A simulation framework makes these relationships observable while keeping experiments repeatable.

1.3 Problem Statement

Design and implement a configurable cloud simulation framework that models a virtualized data centre, executes configurable VM and cloudlet workloads, monitors resource utilization and energy-related metrics, records migration and SLA events, and presents the results through a live dashboard and machine-readable reports. The framework should support different infrastructure scales so that engineers can compare configurations without deploying physical cloud infrastructure.

1.4 Project Aim

The aim is to develop a Green Cloud simulation and monitoring framework that enables energy-aware evaluation of virtualized cloud infrastructure and provides a foundation for intelligent VM placement and energy optimization.

1.5 Project Objectives

- Design a configurable CloudSim Plus simulation environment for hosts, VMs and cloudlets.
- Implement monitoring components that capture host, VM, cloudlet, energy, migration and SLA metrics.
- Develop a dashboard that presents simulation KPIs and time-series information clearly.
- Export simulation metrics to CSV files for reproducible analysis and reporting.
- Validate the framework using small, medium and scaled test configurations.
- Compare the observed results and identify how workload scale affects utilization, energy and migration behaviour.
- Create a modular foundation for future AI-aware VM placement and energy optimization.

1.6 Scope

Included: configurable host and VM counts; CloudSim Plus datacenter simulation; cloudlet workload execution; host and VM resource monitoring; RAM and CPU utilization; energy estimation; migration and SLA event recording; live dashboard visualization; CSV report generation; multiple test configurations; comparative analysis.

Excluded: physical data-centre deployment; measurement from real electrical meters; commercial cloud billing integration; production-grade Kubernetes or container orchestration; real-world carbon accounting; autonomous deployment to public cloud providers. These are outside the current academic scope.

Assumptions: the simulation environment represents infrastructure behaviour sufficiently for comparative experimentation; the configured host CPU capacity and VM requirements are valid for

the selected scenarios; energy is represented through the implemented simulation model; CloudSim Plus event timing is used as the basis for monitoring snapshots.

Boundary conditions: results depend on the workload configuration and CloudSim Plus model parameters. The reported energy values should be interpreted as simulated energy, not physical electricity-meter measurements. Dashboard snapshots can represent a running or completed state depending on when they are captured.

1.7 Expected Engineering Outcome

The expected outcome is a reusable simulation and analysis prototype for green-cloud experimentation. An engineer should be able to enter an infrastructure configuration, run the simulation, observe live behaviour, inspect host-level utilization, review energy accumulation, examine migration activity, and export the resulting data for comparison.

A second outcome is improved visibility. Raw simulation logs are difficult to interpret during demonstrations, especially when many hosts and VMs are involved. The dashboard converts these values into KPI cards, graphs and tables so that the system state can be communicated quickly.

A third outcome is extensibility. The project separates simulation, monitoring, energy modelling, analytics and UI responsibilities. This makes it possible to introduce more advanced placement or optimization algorithms later without redesigning the entire reporting layer.

From an engineering decision-support perspective, the framework can therefore be described as a pre-deployment simulation and evaluation tool: it helps study the likely behaviour of a proposed infrastructure configuration before the corresponding physical resources are purchased or provisioned.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The literature review focused on recent work from 2024–2026 related to green cloud computing, energy-aware VM placement, VM migration, consolidation, CloudSim/CloudSim Plus, and sustainable data-centre resource management. Recent papers were selected because the project aims to align with current research directions rather than older static scheduling approaches.

2.2 Review of Existing Work

Existing Work / Approach	Year	Main Focus	Engineering Relevance / Limitation
Modeling the Green Cloud Continuum: Integrating Energy Considerations into Cloud-Edge Models	2024	Energy factors and green cloud-edge modelling	High-level modelling focus; not an end-to-end live monitoring dashboard.
Cost and Energy Aware Migration Through Dependency Analysis of VM Components in Virtual Cloud Infrastructure	2025	Energy-aware migration and consolidation	Primarily migration-focused rather than a general simulation-monitoring platform.
Energy-efficient Cloud Systems: VM Consolidation with P-Robustness Optimization	2025	Robust optimization and VM consolidation	Optimization is the main focus; infrastructure visualization is secondary.
HAPSO: ACO-initialized, Discretization-aware PSO for Energy- and Carbon-efficient VM Consolidation	2025	ACO + PSO optimization with CloudSim Plus	Algorithmically complex and focused on optimization rather than general monitoring.

Existing Work / Approach	Year	Main Focus	Engineering Relevance / Limitation
Towards Dynamic Virtual Machine Placement Based on Safety Parameters and Resource Utilization Fluctuation	2025	Dynamic placement, energy savings and QoS	Algorithm-specific placement focus.
Quality of Service Driven Energy-efficient Computing to Enhance Sustainable Cloud	2026	QoS and energy objectives for VM management	Focuses on optimization and performance rather than an end-to-end configurable dashboard.

The reviewed studies show a strong research trend toward energy-aware placement, migration and consolidation. The present project complements these approaches by concentrating on the experimental infrastructure needed to configure, observe, visualize and compare cloud scenarios.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Static first-fit placement	Simple, fast and easy to understand.	Can ignore energy, workload dynamics and host consolidation.	Useful as a baseline for simulation experiments.
Heuristic / metaheuristic placement	Can search large VM-host solution spaces and optimize multiple objectives.	Higher computational complexity; algorithm-specific tuning.	Provides future direction for intelligent placement.
Energy-aware migration	Can reduce resource wastage and consolidate VMs.	Migration decisions can be expensive and depend on workload dynamics.	Relevant to migration monitoring.
CloudSim / CloudSim	Repeatable and low-	Results depend on	Core experimental

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Plus experiments	risk environment for cloud experiments.	simulation assumptions and models.	platform used in this project.
Dashboard-driven monitoring	Makes simulation state easy to inspect and communicate.	Visualization alone does not optimize placement.	Main contribution of the monitoring layer.

2.4 Research / Engineering Gap

Recent literature provides sophisticated optimization algorithms, but many studies are designed around a particular placement or migration technique. For an engineering learner or infrastructure planner, there is also a need for a transparent experimental platform in which configuration can be changed easily and the resulting infrastructure behaviour can be observed through consistent metrics.

The identified gap is therefore not that optimization algorithms are unavailable. Instead, the gap addressed by this project is the integration of configurable simulation, continuous monitoring, energy estimation, migration/event tracking, visualization, and CSV-based reporting into one demonstrable framework.

2.5 Proposed Contribution

The project contributes a modular simulation-and-monitoring environment that can serve as the experimental layer for future AI-aware optimization. It emphasizes observability and repeatability: the same workload model can be executed with different host/VM configurations, and the resulting metrics can be compared without changing the physical infrastructure.

2.6 Proposed Contribution and Positioning

The Green Cloud Framework is positioned between a basic cloud simulator and a fully autonomous cloud optimizer. CloudSim Plus provides the simulation engine, while the project adds a configuration layer, monitoring subsystem, energy accounting, event logging, dashboard visualization, and report generation.

The framework is intentionally modular. The simulation core is responsible for creating the datacenter, hosts, VMs and cloudlets and for executing the event-driven simulation. The monitoring subsystem samples resource state and maintains snapshots. The energy component estimates power and accumulated energy using the configured model. The UI layer consumes monitoring data and presents it without changing the simulation logic.

This separation is important because visualization bugs should not change the simulation results. During development, dashboard layout and snapshot-selection issues were corrected while preserving the underlying simulation and CSV generation logic. The final system therefore treats monitoring and presentation as consumers of simulation state rather than as replacements for it.

The project also supports scaling experiments. The demonstrated cases include 10 hosts/20 VMs, 50 hosts/150 VMs, and 50 hosts/200 VMs. The ability to change these inputs makes the framework useful for sensitivity studies. For example, increasing the VM count while keeping host capacity constant can expose utilization changes, migration activity, or SLA pressure.

The framework is not presented as a replacement for production cloud orchestration. Its role is to provide a controlled engineering test bed. A future version can introduce machine-learning predictions, more advanced placement scoring, dynamic migration policies, host power states, renewable-energy inputs, and carbon-intensity data while retaining the existing monitoring architecture.

In practical terms, the contribution can be explained as follows: before buying or provisioning a particular cloud infrastructure configuration, an engineer can simulate representative workloads, inspect resource and energy behaviour, compare alternative configurations, and use the evidence to support the infrastructure decision.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder / User Requirements

Stakeholder	Requirement
Student / Researcher	Configure infrastructure size and workload parameters without editing simulation source code.
Course Faculty	Demonstrate measurable simulation results, monitoring, testing and engineering analysis.
Infrastructure Planner	Compare host/VM configurations using utilization, energy, migration and SLA indicators.
Project Evaluator	View clear dashboard outputs and reproducible test cases.

Functional & Non-Functional Requirements

Requirement	Type	Measurable Target
Configurable hosts and VMs	Functional	Launcher accepts integer host and VM counts.
Workload configuration	Functional	Cloudlet duration and CPU requirements can be entered.
Resource monitoring	Functional	Host and VM CPU/RAM metrics displayed.
Energy reporting	Functional	Energy shown in Wh and exported.
Migration/event monitoring	Functional	Migration count and event

Requirement	Type	Measurable Target
		information recorded.
Dashboard visualization	Functional	At least four time-series/summary views.
CSV export	Functional	Event, host, VM, cloudlet and summary files generated.
Build reliability	Non-Functional	Maven package completes successfully.
Usability	Non-Functional	Simulation can be launched without changing Java code.

3.2 Constraints & Applicable Standards

The applicable constraints are those directly relevant to a software-only simulation prototype. The project does not use physical equipment or production cloud credentials, so hardware safety and live infrastructure control are outside the present boundary.

Standard / Code	Requirement	How Applied in This Project
IEEE 29148	Requirements engineering and specification practices	Functional and non-functional requirements are stated as measurable project requirements.
IEEE 1016	Software design description principles	Architecture, modules, interfaces and system responsibilities are documented.
IEEE 829 / ISO/IEC/IEEE 29119	Software test documentation principles	Test methods, acceptance criteria and verification evidence are organized in Chapter 4.

3.3 Design Specifications

Parameter	Target / Requirement	Unit	Priority
Host count	User configurable hosts	hosts	High
VM count	User configurable VMs	VMs	High
Cloudlet length	User configurable MI	MI	High
Host CPU capacity	User configurable MIPS	MIPS	High
VM CPU requirement	User configurable MIPS	MIPS	High
Energy model	Linear idle-to-maximum model	W / Wh	High
Monitoring interval	Configurable simulated interval	s	High
Dashboard	Live KPIs + graphs + host table	UI	High
Reports	CSV export for core metrics	files	High

3.4 Alternative Solutions & Evaluation

Alternative 1: Basic CloudSim execution. Run the simulation and inspect console output only. This is easy to implement but provides limited visibility and makes comparison difficult.

Alternative 2: Simulation plus monitoring dashboard. Add a structured monitoring layer and visual dashboard. This provides clearer evidence, better demonstrations and reusable exports while retaining the simulation engine.

Alternative 3: Full AI optimization from the beginning. Integrate advanced machine-learning or metaheuristic placement immediately. This offers optimization capability but increases implementation complexity and makes debugging the basic infrastructure harder.

Criterion	Weight	Alternative 1	Alternative 2	Alternative 3
Performance	25%	High	High	Medium
Cost / effort	20%	High	High	Low
Observability	25%	Low	High	Medium
Reliability	20%	Low	High	High
Extensibility	10%	Medium	High	High

Alternative 2 was selected because it balances reliable simulation, measurable outputs and clear demonstration. Advanced AI optimization can be added later.

3.5 Selected Design & Detailed Design

The selected design is a modular simulation-and-monitoring framework in which the simulation engine, monitoring subsystem, energy model, dashboard and CSV reporting layer have distinct responsibilities. This selection is supported by the evaluation in Section 3.4: the dashboard-based approach provides high observability, reliability and extensibility without the additional complexity of immediate full AI optimization.

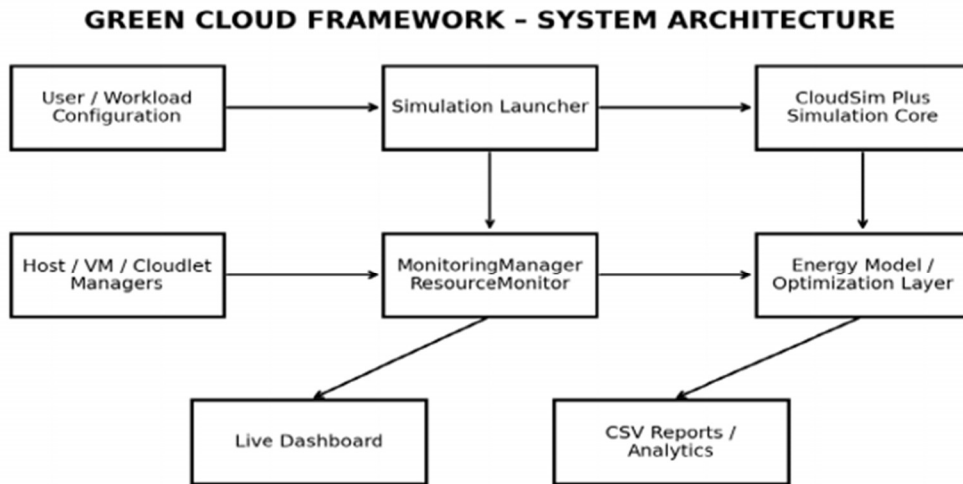


Figure 1. System Architecture

The architecture separates the framework into functional layers. The Simulation Launcher accepts user parameters. The CloudSim Plus core creates the simulated datacenter and executes the event-driven workload. Host, VM and cloudlet managers handle resource entities. The monitoring layer captures runtime state and produces snapshots. The energy and optimization layer evaluates resource state and energy behaviour. Finally, the Live Dashboard and CSV reporting layer present the results.

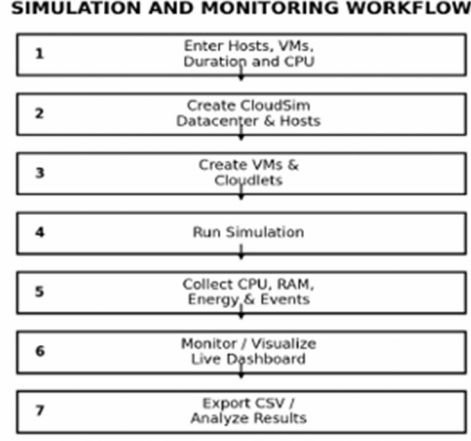


Figure 2. Simulation and Monitoring Workflow

The workflow moves from configuration to simulation, resource creation, execution, monitoring, visualization and export. This separation reduces coupling: a change to the dashboard layout does not require changes to the simulation model, and a future optimization module can consume host metrics without redesigning the report interface.

3.5.1 Detailed Design and Energy Model

The simulation is event-driven. Hosts provide processing capacity and memory resources, VMs consume those resources, and cloudlets represent workload execution. Monitoring snapshots are collected at configured simulation intervals. Each snapshot can contain host metrics, VM metrics, cloudlet metrics, energy information and event information.

The energy component uses a linear power model. A host has an idle power approximately equal to 150 W and a maximum power approximately equal to 300 W. For a utilization fraction u , the conceptual model is:

$$P(u) = P_{\text{idle}} + u \times (P_{\text{max}} - P_{\text{idle}})$$

With $P_{\text{idle}} = 150$ W and $P_{\text{max}} = 300$ W, a host at 0% utilization is represented by 150 W, while a host at 100% utilization is represented by 300 W. Accumulated energy is then obtained from power over simulated time and expressed in Wh.

This model is intentionally simple and transparent. It allows the project to demonstrate the relationship between utilization and energy without claiming the electrical behaviour of a particular commercial server. A more detailed future model could incorporate CPU frequency, memory power, cooling overhead, power states, workload type, or measured server power curves.

Host utilization and VM utilization are related but are not always numerically identical. A VM can be allocated CPU capacity but be idle because its cloudlets have completed or because the workload is not currently consuming CPU. Similarly, a host may have reserved resources while its instantaneous CPU consumption is low. This distinction explains observations such as Host 0 showing 0% CPU while still containing VMs and consuming memory.

The dashboard therefore reports values from authoritative monitoring snapshots rather than fabricating a balanced distribution. Differences among hosts are meaningful evidence of the simulated workload state.

3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Simulation engine	Basic CloudSim execution	Simple and repeatable	Limited observability	Retain CloudSim Plus as the core and add monitoring.
Monitoring	Console-only output	Low implementation effort	Difficult to inspect and compare	Use structured monitoring and dashboard.
Optimization timing	Immediate full AI optimization	Potential optimization capability	Higher complexity and debugging burden	Defer advanced AI to future work after baseline observability is stable.
Energy model	Detailed physical server model	Higher physical realism	Requires measured data and more assumptions	Use transparent linear model for controlled comparison.

3.7 Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Energy should be treated as a first-class infrastructure metric.	Energy vs Time graph; Wh totals; host power model.	Energy is included in monitoring and comparison rather than treated as an afterthought.
Ethics	Results must distinguish simulation from physical	CloudSim-based methodology and	Report states that Wh values are simulated

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
	measurement.	explicit assumptions.	estimates.
Health & Safety	No physical electrical or mechanical equipment is used.	Software-only simulation.	Physical equipment risk is avoided during experimentation.
Security	Reports and configuration should not require sensitive production data.	Synthetic/configurable workloads and local CSV output.	No production credentials or cloud secrets are required.

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Dashboard displays stale/cleanup snapshot	Medium	Medium	Medium	Track workload snapshots separately from final cleanup state	Low
Simulation parameters produce unrealistic utilization	Medium	Medium	Medium	Use multiple test cases and document assumptions.	Low
Large configurations increase runtime overhead	Medium	Medium	Medium	Scale gradually and monitor resource use.	Low
Interpretation of simulated energy as physical power	Low	High	Medium	Explicitly label energy as simulated and model-based.	Low

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Report/data file inconsistency	Low	Medium	Low	Use fixed CSV outputs and verification runs.	Low

Security is primarily a data-integrity and software-use concern in the present prototype. Because the system does not connect to a production cloud provider, the attack surface is smaller than that of a live infrastructure controller. Future versions that integrate cloud APIs would require authentication, authorization, secret management, audit logs and secure transport.

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

Development followed an incremental modular approach. The first stage established the CloudSim Plus simulation core and verified that hosts, VMs and cloudlets could be created and executed. The second stage added monitoring and CSV exports. The third stage refined the dashboard so that monitoring snapshots were presented correctly, including protection against cleanup snapshots overwriting the final workload state. The project was then validated with multiple scale configurations.

The implementation was built using Java 17 and Maven. CloudSim Plus provides the simulation entities and event-driven execution. SLF4J and Logback provide structured application logging. The graphical dashboard presents KPI cards, charts and tables, while CSV files provide a simple interoperable format for analysis.

Tool / Software / Equipment	Purpose	Stage Used
Java 17	Primary implementation language	All stages
Maven	Build, dependency management and packaging	All stages
CloudSim Plus 8.5.x	Cloud infrastructure simulation	Simulation core
SLF4J	Logging abstraction	Runtime logging
Logback	Logging implementation	Runtime logging
Swing / Java UI components	Launcher and dashboard interface	Monitoring/UI
CSV	Exported measurement/report format	Monitoring/reporting
IntelliJ IDEA	Development, debugging and project integration	All stages

The project is software-only; no physical cloud hardware, sensors or fabrication equipment are required.

4.2 Hardware / Software / Fabrication Implementation & Modern Tools Used

4.2.1 Simulation Setup

CloudSim Plus is used as the central simulation engine. The application creates a simulated datacenter containing a configurable number of hosts. Each host has CPU and memory resources, and VMs are created with configurable CPU requirements. Cloudlets are then submitted to execute the workload. The event-driven simulator advances through simulation time and invokes the monitoring logic at configured intervals.

The high-level setup sequence is: initialize the CloudSim simulation; create the datacenter; create hosts; create the broker; create VMs; create cloudlets; submit VMs and cloudlets; attach monitoring; start the simulation; collect snapshots; finish execution; and generate summary reports.

The modular implementation uses manager classes so that the responsibilities are understandable and independently testable. Typical responsibilities include datacenter creation, host creation, VM creation, cloudlet creation, monitoring, energy calculation, event logging and dashboard rendering.

4.2.2 Configuration Inputs

Parameter	Case 1	Case 2	Case 3
Hosts	10	50	50
VMs	20	150	200
Cloudlet duration / length	10000	50000	50000
Host CPU capacity	10000 MIPS	10000 MIPS	10000 MIPS
VM CPU requirement	2500 MIPS	2500 MIPS	2500 MIPS

The launcher makes these values visible to the user so that test scenarios can be changed without editing the source code. This improves repeatability and supports demonstration of sensitivity to infrastructure scale.

4.2.3 Monitoring Subsystem

The monitoring subsystem is responsible for converting CloudSim state into structured measurements. ResourceMonitor reads host and VM resource utilization from the simulation objects. MonitoringManager coordinates snapshot creation. EventLogger records significant simulation events. PerformanceCalculator derives summary metrics, while exporters write the collected data to CSV files.

The exported files include event_log.csv, host_metrics.csv, vm_metrics.csv, cloudlet_metrics.csv, and simulation_summary.csv. Keeping these outputs stable allows the dashboard and external analysis tools to consume the same data.

4.2.4 Dashboard

The Live Dashboard presents two rows of KPI cards: current energy, average host CPU, average VM CPU, active hosts, running VMs, running cloudlets, completed cloudlets, SLA violations, migrations, and simulation time. Below the cards, four graphs show energy accumulation, host utilization, VM utilization, and migration events. A host summary table provides CPU, RAM, power, energy, VM count and status for each host.

A significant implementation detail is snapshot selection. A final cleanup event can occur after the workload has completed and may report zero active CPU or zero running VMs. If that cleanup snapshot were used as the dashboard's workload state, valid utilization information would disappear. The dashboard therefore retains the latest meaningful workload snapshot for CPU, VM and cloudlet KPIs while using authoritative final values for time, energy, migration and SLA totals.

This design preserves correctness without changing the simulation algorithm. It also explains why a host may display 0% CPU while still containing VMs: the dashboard is reporting the actual instantaneous CPU value from the selected snapshot, not the amount of reserved CPU capacity.

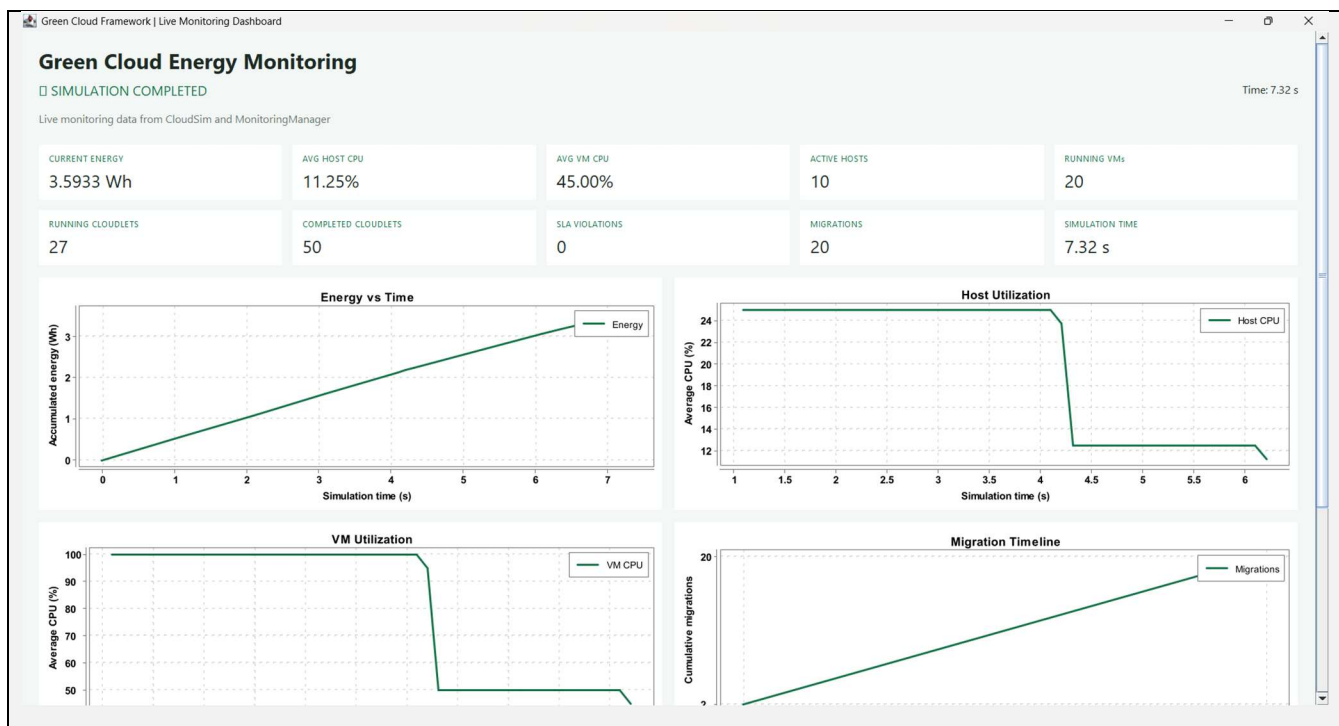


Figure 6. Representative Live Monitoring Dashboard

4.3 Test Plan & Verification

Testing was performed by changing the launcher inputs and executing the same simulation workflow. The objective was to verify configurability, completion, monitoring, energy reporting, migration tracking, dashboard rendering, and scaling behaviour.

Requirement	Test Method	Acceptance Criterion
Configurable infrastructure	Run multiple host/VM combinations	Launcher accepts values and simulation starts.
Simulation completion	Execute each scenario to completion or capture stable running state	Dashboard reports simulation time and workload state.
Energy monitoring	Compare dashboard energy with exported summary	Energy is displayed in Wh and is non-negative.
Host monitoring	Inspect host table and host graph	Host CPU/RAM/power values are displayed.
VM monitoring	Inspect VM KPI and VM graph	VM CPU utilization is displayed.
Migration tracking	Inspect migration KPI and timeline	Migration count is reported.
SLA monitoring	Inspect SLA KPI and logs	SLA violations are visible when generated.
CSV reporting	Inspect generated files	Expected CSV files are created.
Scale test	Run 10/20 and 50/150 and 50/200 scenarios	Application remains operational and produces metrics.

Verification also included a build check using Maven. The project was packaged successfully after the dashboard refinements. The tests were designed to detect both functional errors and misleading visualization states.

The test methodology intentionally avoids claiming that one configuration is universally optimal. Instead, each test case is treated as evidence about the behaviour of that specific simulated scenario.

4.4 Results

Metric	Case 1	Case 2	Case 3
Hosts	10	50	50
VMs	20	150	200
Cloudlet duration / length	10000	50000	50000

Metric	Case 1	Case 2	Case 3
Host CPU capacity	10000 MIPS	10000 MIPS	10000 MIPS
VM CPU requirement	2500 MIPS	2500 MIPS	2500 MIPS
Energy	3.5933 Wh	45.9306 Wh	43.4115 Wh
Average Host CPU	11.25%	4.08%	3.06%
Average VM CPU	45.00%	16.33%	12.25%
Active Hosts	10	50	50
Running VMs	20	150	200
Running Cloudlets	27	49	49
Completed Cloudlets	50	50	50
SLA Violations	0	0	3450
Migrations	20	150	200
Simulation Time	7.32 s	21.21 s	20.21 s

Case 1 represents the smaller configuration. The dashboard showed 3.5933 Wh of simulated energy, 11.25% average host CPU utilization and 45% average VM CPU utilization. All 20 VMs were represented, 50 cloudlets were completed, and the reported SLA violation count was zero.

Case 2 increases the infrastructure to 50 hosts and 150 VMs. The larger configuration produced a higher total energy value because more infrastructure was active over a longer simulation period. At the same time, average host CPU utilization was lower because the workload was distributed across many hosts. VM CPU utilization was also lower than in Case 1.

The migration count scaled with the number of VMs in these experiments. This is useful for demonstration because it shows that migration activity is an observable infrastructure event rather than a hidden internal process.

4.4.1 Case 3 and Cross-Case Analysis

Case 3 used 50 hosts and 200 VMs with a cloudlet length of 50000, host capacity of 10000 MIPS and VM requirement of 2500 MIPS. The captured dashboard state reported 43.4115 Wh energy, 3.06% average host CPU, 12.25% average VM CPU, 50 active hosts, 200 running VMs, 49 running cloudlets, 50 completed cloudlets, 3450 SLA violations and 200 migrations at approximately 20.21 s.

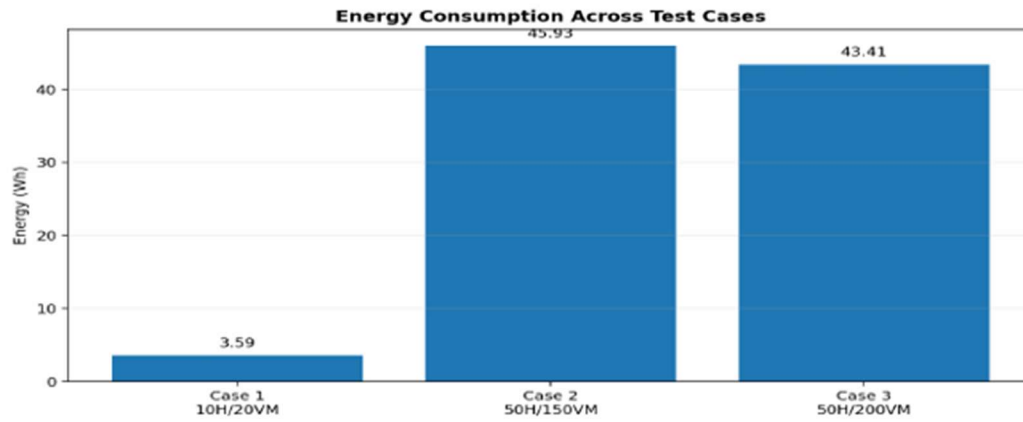


Figure 3. Energy consumption across the three demonstrated test cases.

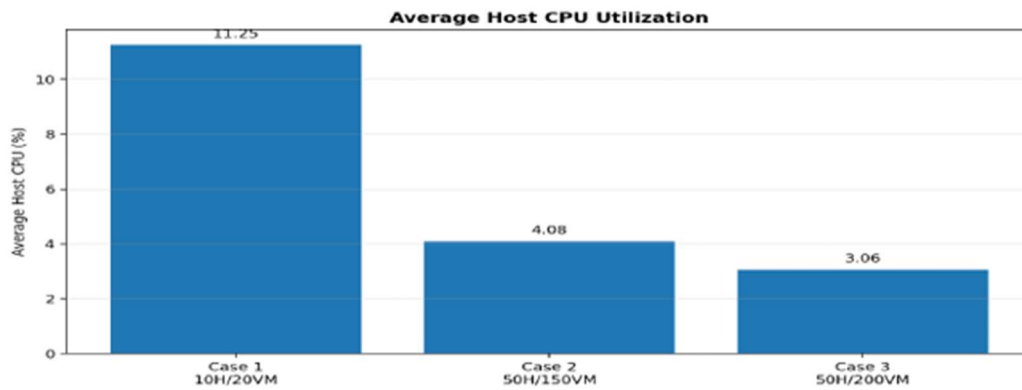


Figure 4. Average host CPU utilization across the three demonstrated test cases.

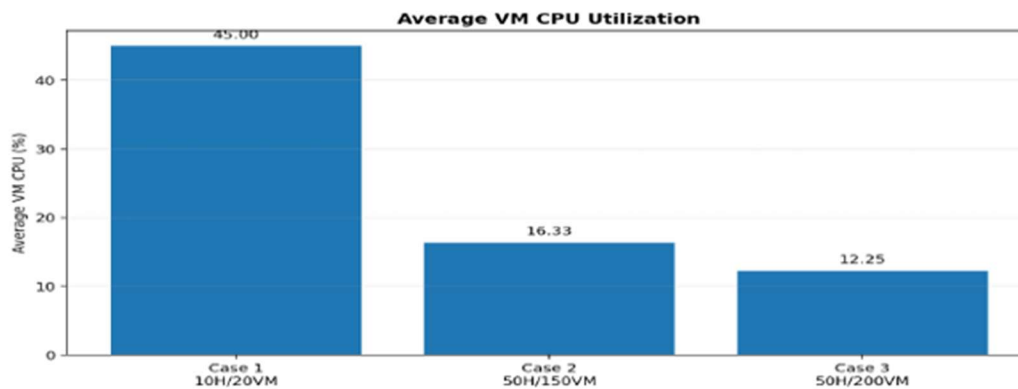


Figure 5. Average VM CPU utilization across the three demonstrated test cases.

4.5 Data Analysis & Engineering Judgment

The graphs show an important distinction between workload density and infrastructure scale. The smaller Case 1 produces higher average utilization because fewer hosts share the workload. Cases 2 and 3 use 50 hosts, so average host utilization is lower even though the VM population is much larger. This demonstrates why host count alone cannot be used to judge efficiency.

The migration timeline provides another dimension. Migration count increases from 20 to 150 to 200 across the three configurations. The dashboard therefore exposes both resource utilization and infrastructure-management activity.

Case 3 also demonstrates the importance of workload-state interpretation. Its captured running state reports substantially more SLA violations than Cases 1 and 2. Because the value is tied to the captured simulated state and configuration, it should be treated as scenario-specific evidence rather than as a universal performance limit.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement (Fully / Partially / Not Achieved)
Configurable CloudSim environment	Launcher accepts hosts, VMs, workload and CPU inputs.	Fully achieved
Monitoring subsystem	Host/VM/cloudlet snapshots and KPIs.	Fully achieved
Energy estimation	Wh values and Energy vs Time graph.	Fully achieved
Dashboard visualization	Four graphs, KPI cards and host table.	Fully achieved
CSV reporting	Event, host, VM, cloudlet and summary files.	Fully achieved
Multi-scale validation	10/20, 50/150 and 50/200 scenarios.	Fully achieved
AI-aware optimization	Architecture prepared for future extension.	Partially achieved / future work

4.7 Strengths & Limitations

Strengths:

- Configurability of host, VM and workload inputs.
- Modular separation of simulation, monitoring, energy and presentation.
- Repeatable test configurations and machine-readable CSV reports.
- Clear visualization of KPIs, resource utilization and migration activity.
- Low-cost software-only experimental approach.

Limitations:

- Dependence on simulation assumptions and CloudSim Plus model parameters.
- Simplified linear energy model rather than physical power measurements.
- No physical cloud deployment or electrical-meter validation.
- Advanced AI-aware placement is prepared architecturally but not fully implemented in the demonstrated results.
- Dashboard snapshots depend on meaningful workload-state selection.

The present version should therefore be described as a simulation and decision-support prototype rather than a production cloud controller.

4.8 Project Planning, Roles & Budget

Milestone	Activity	Output
M1	Problem definition and literature review	Project scope and research gap
M2	CloudSim Plus simulation core	Working datacenter, hosts, VMs and cloudlets
M3	Monitoring subsystem	Host/VM/cloudlet metrics and event logging
M4	Energy and reporting	Energy model and CSV exports
M5	Dashboard	KPI cards, graphs and host table
M6	Testing and debugging	Multiple scale configurations verified
M7	Documentation and presentation	Final report, results and demonstration

Student	Assigned Responsibility	Actual Contribution
Deepak R (192511008)	Simulation integration, monitoring, dashboard, reporting	Implemented and integrated major software modules; performed debugging and validation.

Student	Assigned Responsibility	Actual Contribution
Bharath S (192521034)	Experiment design, validation support, literature and documentation	Supported test configuration, result analysis, literature review and report preparation.

Item	Estimated Cost	Actual Cost
Development software	₹0 – academic/open-source tools	₹0
CloudSim Plus and Java tooling	₹0 – open-source / academic use	₹0
Hardware for prototype	Not required	₹0
Cloud infrastructure	Not required for current simulation	₹0
Printing / documentation	Institution dependent	Not included

The software-only approach keeps the prototype cost low and avoids cloud-service charges during development. This is one of the main advantages of simulation for early-stage infrastructure studies.

CHAPTER 5

CONCLUSION & REFLECTION

5.1 Conclusion

The Green Cloud Framework successfully demonstrates how a configurable cloud simulation can be used to study infrastructure behaviour before physical deployment. CloudSim Plus provides the simulated data-centre environment, while the monitoring layer captures resource and event information and the dashboard converts these measurements into understandable engineering evidence.

The three demonstrated test configurations show that infrastructure scale strongly influences utilization and energy behaviour. The smaller 10-host/20-VM configuration produced higher average utilization, while the 50-host configurations distributed the workload more widely. Energy and migration metrics provide additional evidence for evaluating the consequences of a chosen configuration.

The project also establishes a useful distinction between simulation state and physical infrastructure measurements. The reported energy values are model-based simulation outputs. They are valuable for comparison within the same modelling assumptions, but they should not be interpreted as direct measurements of electricity consumption in a commercial data centre.

5.2 Future Work

- Integrate AI-based VM placement and workload-aware host scoring.
- Implement dynamic migration, consolidation and simulated host power states.
- Compare baseline, energy-aware and AI-aware placement strategies.
- Add renewable-energy, carbon-intensity and richer measured power models.
- Extend analytics with historical runs and statistical comparison.

5.3 Professional Learning & Individual Reflection

Deepak R: The main challenge was synchronizing the dashboard with the correct simulation snapshot. This strengthened my understanding of event-driven simulation, monitoring, CSV reporting and GUI rendering.

Bharath S: The project developed practical skills in repeatable cloud experiments, workload comparison, literature review, result interpretation and technical documentation.

5.3.1 New Knowledge Acquired

- Event-driven cloud simulation using CloudSim Plus.
- Resource monitoring and snapshot-based metric collection.
- Energy estimation using a transparent utilization-based model.
- CSV-based reporting and dashboard data handling.
- Interpretation of utilization, migration and SLA behaviour across scaled scenarios.

5.3.2 Engineering & Professional Skills Developed

- Modular software architecture and integration.
- Experiment planning and repeatable verification.
- Engineering judgment from simulation evidence.
- Technical documentation and visualization.
- Team collaboration, debugging and presentation.

5.3.3 Individual Reflection

Deepak R: The most difficult engineering problem was ensuring that the dashboard used the correct workload snapshot rather than a final cleanup snapshot. The solution was to retain the latest meaningful workload state for utilization-related KPIs while using authoritative final values for cumulative metrics. A key decision was to keep monitoring separate from the simulation core because this reduced coupling and made debugging safer. If repeated, I would extend the monitoring model with richer historical state and automated validation.

Bharath S: A key challenge was comparing multiple configurations without changing the experimental workflow. I supported a repeatable configuration-based approach and used the resulting metrics for comparison rather than relying on a single run. The project strengthened my knowledge of cloud resource management, literature analysis and technical reporting. If repeated, I would add more test configurations and statistical comparison across repeated runs.

REFERENCES

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- [4] “HAPSO: An ACO-initialized, discretization-aware PSO for energy- and carbon-efficient VM consolidation in green cloud datacenters,” *Sustainable Computing: Informatics and Systems*, vol. 48, art. 101258, 2025. DOI: 10.1016/j.suscom.2025.101258.
- [5] “Towards dynamic virtual machine placement based on safety parameters and resource utilization fluctuation for energy savings and QoS improvement in cloud computing,” *Future Generation Computer Systems*, vol. 171, art. 107853, 2025. DOI: 10.1016/j.future.2025.107853.
- [6] “Quality of service driven energy-efficient computing to enhance sustainable cloud,” *Array*, art. 100790, 2026. DOI: 10.1016/j.array.2026.100790.

The report uses IEEE-style references. Literature, software frameworks, libraries, standards, and other external resources used during the project should be acknowledged consistently. The source report provided the six references above; no additional references have been invented in this conversion.

APPENDICES

Appendix A – Detailed Calculations

For utilization fraction u , the simulated host power model is $P(u)=150+150u$ watts. Energy is accumulated over simulated time and reported in Wh.

At $u = 0$, $P(0) = 150$ W. At $u = 1$, $P(1) = 300$ W. For intermediate utilization, power is linearly interpolated between these limits. The detailed derivation is intentionally kept compact in the main report because the purpose of the model is transparent comparative simulation rather than physical server characterization.

Appendix B – Engineering Drawings / Schematics

The engineering architecture and operational workflow are provided in Chapter 3 as Figures 1 and 2. These diagrams represent the system-level interfaces among configuration, simulation, monitoring, energy evaluation, dashboard visualization and reporting.

Appendix C – Source Code / Code Repository Information

The complete source code is maintained separately in the project workspace. The report documents the architecture, interfaces, implementation approach, test plan and outputs rather than reproducing the entire Java source code.

Appendix D – Datasheets

Not applicable to the software-only simulation prototype. No physical sensors, actuators or fabrication components were used.

Appendix E – Experimental / Raw Data

Primary configurations: Case 1 = 10 hosts / 20 VMs; Case 2 = 50 / 150; Case 3 = 50 / 200, with the workload and CPU settings documented in Chapter 4.

Metric	Case 1	Case 2	Case 3
Energy (Wh)	3.5933	45.9306	43.4115
Average Host CPU (%)	11.25	4.08	3.06
Average VM CPU (%)	45.00	16.33	12.25
Migrations	20	150	200
SLA Violations	0	0	3450

Appendix F – Ethics / Institutional Approval

The project is a software-only academic simulation and does not use human subjects, personal datasets, or physical experimental equipment. Ethical handling therefore focuses on accurate reporting, acknowledgement of external resources, and clear distinction between simulated and measured quantities.

Appendix G – Project Meeting / Progress Records

The development milestones are summarized in Section 4.8. Detailed meeting records can be maintained separately if required by the department.

Appendix H – User / Industry Feedback

No formal industry feedback dataset was included in the source report.

Appendix I – Publications / Patents / Competitions / Product Outcomes

No publication, patent, competition result, or commercial product outcome was included in the source report.

Appendix J – Individual Contribution Evidence

The mandatory contribution statement is provided at the beginning of the report and summarized again in Section 4.8. Both members participated in discussions, integration decisions, verification and final demonstration preparation.

CAPSTONE PROJECT OUTCOME MAPPING SHEET

To be completed / reviewed by the department or faculty assessor.

Outcome	Evidence	SO / Area	Location in This Report
Complex engineering problem formulation	Problem definition, interacting parameters and engineering gap	SO1	Ch. 1, Ch. 2
Engineering design under constraints	Requirements, alternatives, architecture, trade-offs and risks	SO2	Ch. 3
Written / oral technical communication	Structured report, figures, tables and presentation	SO3	Whole report + Viva
Ethics and professional responsibility	Simulation limitations, acknowledgement and integrity	SO4	Ch. 3 (3.7)
Teamwork and leadership	Individual contribution, roles and project planning	SO5	Ch. 4 (4.8)
Experimentation, analysis, engineering judgment	Test plan, results, comparison and interpretation	SO6	Ch. 4 (4.3–4.6)
Independent acquisition of new knowledge	Professional learning and individual reflection	SO7	Ch. 5 (5.3)
Standards	Requirements, design and test documentation practices	SO2	Ch. 3 (3.2)
Sustainability and societal / environmental impact	Energy monitoring and software-only sustainable approach	SO2/SO4	Ch. 3 (3.7)

Outcome	Evidence	SO / Area	Location in This Report
Risk assessment and mitigation	Risk register and mitigation actions	SO2/SO4	Ch. 3 (3.7)
Security considerations	Data integrity and absence of production credentials	Relevant SO	Ch. 3 (3.7)
Integrated / system-level engineering	Layered architecture and subsystem interfaces	SO1/SO2	Ch. 3 (3.5)
Project planning and management	Milestones, roles and budget	SO5	Ch. 4 (4.8)
Individual contribution	Contribution statement and project role evidence	SO1–SO7	Contribution Statement + Ch. 4 (4.8)

Note: The course code/name and supervisor field on the cover are intentionally marked for confirmation because the source report did not provide a separate course code/name or an explicit supervisor designation; the course faculty name was provided and has been carried into the supervision field for editing if needed.